

NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2017-2018

CE1005/CZ1005 – DIGITAL LOGIC

Apr/May 2018

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. All questions carry equal marks.

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1. (a) A 16-bit value 810C (hex) is stored in a digital system. Determine the **decimal value** (including its sign where applicable) it represents in each case below. Show all steps clearly.
 - (i) Unsigned (2 marks)
 - (ii) Sign-magnitude (2 marks)
 - (iii) Two's complement (2 marks)
 - (b) Some temperature values are to be stored in a digital system. The temperature range is 0.0 to 150.0 degrees Celsius, recorded to the nearest 0.1 degree. For example, a temperature may be recorded as 149.7 degrees Celsius. Each recorded temperature is to be stored as an unsigned fixed-point binary value. Determine the **minimum number of bits** that must be set aside to store each value. Show all steps clearly. (3 marks)

Note: Question No. 1 continues on Page 2

- (c) Simplify the following Boolean expression algebraically. Show all steps clearly and give your answer in SOP (sum-of-product) form.

$$F = [(u + v w') (u' x + w)' + x (u' v')']'$$

(8 marks)

- (d) (i) Perform the following 8-bit 2's complement arithmetic operations and state whether or not there is an overflow in each case.

- 49 (hex) + B6 (hex)
- F3 (hex) – 7A (hex)

(4 marks)

- (ii) Briefly describe the meaning of arithmetic overflow and how it may be detected in a 2's complement add/subtract circuit.

(4 marks)

2. (a) Table Q2 shows the truth table of a combinational logic circuit with active-low inputs and active-high outputs.

Table Q2

Inputs								Outputs		
E*	A7*	A6*	A5*	A4*	A3*	A2*	A1*	Y2	Y1	Y0
1	x	x	x	x	x	x	x	0	0	0
0	1	1	1	1	1	1	1	0	0	0
0	1	1	1	1	1	1	0	0	0	1
0	1	1	1	1	1	0	x	0	1	0
0	1	1	1	1	0	x	x	0	1	1
0	1	1	1	0	x	x	x	1	0	0
0	1	1	0	x	x	x	x	1	0	1
0	1	0	x	x	x	x	x	1	1	0
0	0	x	x	x	x	x	x	1	1	1

x denotes "don't care"

- (i) Sketch a logic circuit diagram with suitable logic symbols to illustrate clearly how the output Y1 is asserted.

(5 marks)

- (ii) Briefly describe the purpose of input E*.

(3 marks)

Note: Question No. 2 continues on Page 3

- (b) A circuit is described by the canonical product-of-maxterm expression $F(a,b,c,d)$ together with don't care outputs $X(a,b,c,d)$ below. Use a Karnaugh map to obtain a **minimum-cost POS** (product-of-sum) expression for this circuit. Show the loops clearly.

$$F(a, b, c, d) = \prod M(3, 5, 6, 11, 14)$$

$$X(a, b, c, d) = \prod M(7, 8, 10)$$

(6 marks)

- (c) Figure Q2 shows a circuit that has seven inputs: A, B, C, D, E, F, G and the output is X.

- (i) Identify the characteristic of the logic gate outputs that allows them to be connected together in this manner.

(1 mark)

- (ii) Briefly describe how the circuit works.

(4 marks)

- (iii) Write a Boolean expression for X.

(2 marks)

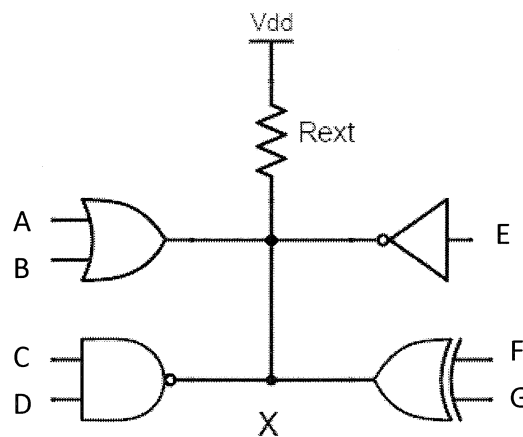


Figure Q2

- (d) Briefly describe each of the following parameters with suitable timing waveforms:

- t_r
- t_f
- t_{PHL}
- t_{PLH}

(4 marks)

3. (a) State True or False for each of the following statements.
- Modules may contain code to describe the hardware but not instance of other modules.
 - Both wire and reg can be assigned outside of always blocks.
 - Statement order matters in a combinational always block.
 - An n-input mux requires $\log_2(n)$ select bits.
 - Parameter is a constant that is local to a module.
- (5 marks)
- (b) Given the sum-of-minterm expression $F(x, y, z) = \sum m(1, 6)$
- (i) Draw the schematic diagram showing the implementation of F using one 2x1 multiplexer, one AND gate and one NOR gate.
- (5 marks)
- (ii) Write the conditional assignment statement to implement F.
- (3 marks)
- (c) You have been hired as a design engineer in a chip manufacturing company to implement an arithmetic logic unit (ALU) for a computing system. The ALU has three inputs *A*, *B* and *OP*. Inputs *A* and *B* are two operands on which the ALU will perform the following arithmetic and logic operations: addition, subtraction, AND, OR and XOR. Input *OP* will determine the operation to be performed on *A* and *B*. The ALU generates a 32-bit output *ALU_OUT*. Based on the number of operations and *ALU_OUT* bit width, determine the correct bit size of *A*, *B* and *OP*. Write a behavioral Verilog module to implement the ALU.
- (12 marks)
4. (a) Differentiate between a level-sensitive D-latch and edge-triggered D-flip-flop with the help of a timing diagram.
- (5 marks)

Note: Question No. 4 continues on Page 5

- (b) A 3-bit odd/even up-counter has a 1-bit input In and a 3-bit output $Count$. When In is 0 it counts the next even value, and when In is 1 it counts the next odd value. The counter goes back to zero (initial state) only when it reaches the maximum odd count. It stays at the maximum even count state when In is 0.
- (i) Draw the state transition diagram for the 3-bit odd/even up-counter. (7 marks)
- (ii) Determine the output $Count$ for the following input sequence: 1,1,0,1,0,1,0. (3 marks)
- (iii) Write the behavioral Verilog module for the odd/even up-counter discussed above using a CASE statement. Assume two additional inputs CLK and $RESET$ for the sequential circuit implementation. (10 marks)

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.